

9.6 Classwork: Geometry Review (Non-calculator)

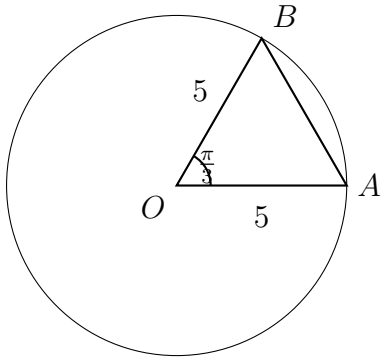
Formula sheet reference (Topics 3.2–3.3):

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \qquad c^2 = a^2 + b^2 - 2ab \cos C \qquad \text{Area} = \frac{1}{2}ab \sin C$$

$$l = r\theta \quad A = \frac{1}{2}r^2\theta$$

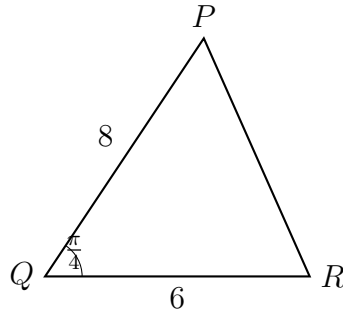
Exact values: $\cos \frac{\pi}{3} = \frac{1}{2}, \quad \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$

1. A triangle OAB is inscribed in a circle with centre O and radius 5. The angle $A\hat{O}B = \frac{\pi}{3}$ radians, and $OA = OB = 5$.



- Find the length of the arc AB . [2]
- Find the area of the sector OAB . [2]
- Using the cosine rule, find the length of AB . [3]
- Find the area of triangle OAB . [2]
- Hence, find the area of the segment (the region between the chord AB and the arc AB). [2]

2. The following diagram shows triangle PQR , where $PQ = 8$, $QR = 6$, and $\angle PQR = \frac{\pi}{4}$ radians.



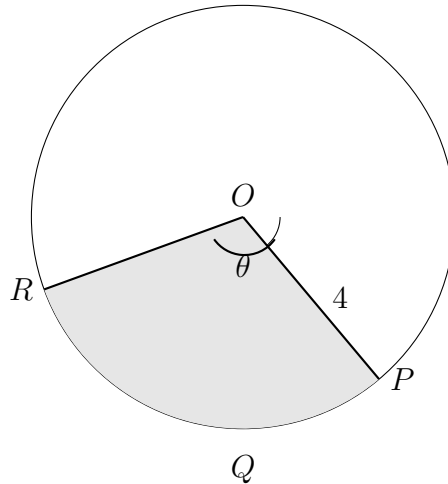
Exact values: $\cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}, \quad \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$

- (a) Find the exact area of triangle PQR . [2]

- (b) Find the exact length of PR . [3]

- (c) Using the sine rule, find $\sin(\hat{R})$. Leave your answer in exact form. [3]

3. A circle has centre O and radius 4 cm. Points P , Q and R lie on the circumference, and $\angle POR = \theta$ radians. The arc PQR has length 10 cm.



- Find the perimeter of the shaded sector. [2]
- Find θ . [2]
- Find the area of the shaded sector. [2]

(IB AA SL, 2023 May P1 TZ2, Q1)

4. The following diagram shows triangle ABC , with $AB = 10$, $BC = x$, and $AC = 2x$.

Given that $\cos \hat{C} = \frac{3}{4}$, find the area of the triangle.

Give your answer in the form $\frac{p\sqrt{q}}{2}$ where $p, q \in \mathbb{Z}^+$. [7]

(IB AA SL, 2021 May P1 TZ2, Q6)

5. Consider triangle ABC , where $AC = 12$, $CB = 7$, and $\hat{BAC} = 25^\circ$.

Find the smallest possible perimeter of triangle ABC .

[5]

(IB AA SL, 2021 Nov P2, Q3) — calculator allowed

6. The cities Lucknow (L), Jaipur (J), and Delhi (D) are represented in the following diagram. Lucknow lies 500 km directly east of Jaipur, and $\hat{J}LD = 25^\circ$. The bearing of D from J is 034° .

(a) Find $J\hat{D}L$. [2]

(b) Find the distance between Lucknow and Delhi. [3]

(IB AA SL, 2023 May P2 TZ1, Q1) — calculator allowed

7. A farmer is placing posts at points A , B , and C in the ground to mark the boundaries of a triangular piece of land on his property.

From point A , he walks due west 230 metres to point B . From point B , he walks 175 metres on a bearing of 063° to reach point C .

- (a) Find the distance from point A to point C . [4]
- (b) Find the area of this piece of land. [2]
- (c) Find $\hat{CAB}$. [3]
- (d) The farmer wants to divide the piece of land into two sections. He will put a post at point D , which is between A and C . He wants the boundary BD to divide the piece of land such that the sections have equal area. Find the distance from point B to point D . [5]

(IB AA SL, 2023 May P2 TZ2, Q7) — calculator allowed

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